QUANTITATIVE RESEARCH PORTFOLIO

From Three Prices to One Hedge: Equity Derivatives on SPX

Three-Way Pricing Cross-Validation → SPX Calibration
Exotic Model Risk → Hedging-Error Attribution

Fourier, PDE and Monte Carlo engines agree on every price; real SPX quotes then decide how much the choice of model matters for calibration, for an autocallable, and for the hedge.

2.00	14.0–16.0%	91%
CRANK–NICOLSON ORDER	AUTOCALL COUPON ACROSS MODELS	HEDGED P&L VARIANCE EXPLAINED

Kexin (Eve) Zhang

Independent Research Project / October 2026

kexinzhang0209@gmail.com

SPX end-of-day option quotes (optionsDX) • 4.93 million quotes • 311 quote dates • Jan–Mar 2020 and Jan–Dec 2023

Coupon range: Black-Scholes, Heston and Dupire local vol, all fitted to the 2023-01-18 surface. P&L explained: daily variance, 2020 Q1 and 2023.

Independent research demonstration. Not investment advice.

Executive Snapshot

THE RESEARCH QUESTION

> **Do three independent numerical methods produce the same price, and once a model is fitted to real SPX options, how much do its remaining degrees of freedom move exotic prices and hedging P&L?**

EVIDENCE IN FOUR NUMBERS

2.00	0.89 vol pt	+2.0 pts	91%
PDE CONVERGENCE ORDER	HESTON SPX FIT RMSE	AUTOCALL COUPON, BS TO LV	P&L VARIANCE EXPLAINED

WHAT THE EVIDENCE SAYS

- Closed form, COS, Crank–Nicolson and Monte Carlo agree on a Black-Scholes call to four decimals (12.0056); Heston COS and QE Monte Carlo agree within 0.2 standard errors.
- Convergence rates are measured, not assumed: PDE slope -2.00, exponential COS convergence, Monte Carlo error $\propto n^{-1/2}$ with an 8$\times$ variance cut from a control variate.
- On the SPX surface the Heston calibration is stable in ρ (CV 3%), but κ and ξ move together (weekly-change correlation 0.96): the ridge predicted beforehand on synthetic data.
- Three models fitted to the same surface give a 3-year autocallable a fair coupon of 14.0%, 15.2% and 16.0%.

WHAT REQUIRES CAUTION

- Heston misses the 1-month skew (1.48 vol pt error); no single diffusive model fits the whole surface inside the bid–ask spread.
- The real hedging backtest covers 15 monthly trades over two windows. It shows the *source* of P&L, not a statistically reliable strategy return.
- The March 2020 trade is truncated at the data cut-off, so the 2020 Q1 loss is a lower bound.
- No 2D Heston PDE and no stochastic local vol yet; quotes are end-of-day mids from one vendor.

WHAT THE PROJECT DEMONSTRATES

A pricing library whose every number is checked by a second, independent method.

That discipline is what makes the later claims believable: when Heston and local vol disagree on an autocallable by 0.8 coupon points, the gap is model risk, not numerical error.

Data: SPX end-of-day quotes, optionsDX, Jan–Mar 2020 and Jan–Dec 2023. Showcase date 2023-01-18 (spot 3945.1). All engines and tests run on synthetic inputs; 70 unit tests run in continuous integration.

1 Research Architecture

1.1 Research sequence

The project is built in four stages, each of which reuses only what the previous stage has already validated. A pricing error in stage 1 would contaminate every later number, so the order is deliberate: numerics first, market data second, products third, risk last.

1.1.1 Questions

1. **Numerics:** do Fourier, finite-difference and simulation methods give the same price for the same contract, and does each converge at its theoretical rate (Fang and Oosterlee, 2008; Andersen, 2008; Longstaff and Schwartz, 2001)?
2. **Calibration:** how well do Heston (Heston, 1993), SABR (Hagan et al., 2002) and Dupire local vol (Dupire, 1994) fit the SPX surface, and which parameters are actually identified by it?
3. **Model risk:** if three models reproduce the same vanilla market, how far apart are their prices for path-dependent products (a barrier put and an autocallable)?
4. **Hedging:** where does the P&L of a delta-hedged option come from, and how much of it can the Greeks explain on real data?

Each stage has a validation step that does not depend on the market: a closed form, a limiting case, or a synthetic surface whose true parameters are known. Real data enter only after the method has passed on synthetic data. That separation is what allows a fit error on SPX to be read as model error rather than as a bug.

1.2 Data and cleaning

End-of-day SPX option quotes (16:00 ET) were taken from optionsDX for January–March 2020 (the COVID crash) and all of 2023 (a calm, rising market). Each row carries bid and ask for the call and the put at one strike and expiry, together with the index level. A single loader maps optionsDX, OptionMetrics and Yahoo snapshots to one schema, so the cleaning and calibration code never sees vendor formats.

Filters, applied per quote date and expiry:

- two-sided quotes only: positive bid, ask above bid, mid of at least 0.05;
- relative spread $(\text{ask} - \text{bid})/\text{mid} \le 50\%$, and at least 7 days to expiry;
- standardised log-moneyness $\ln(K/F)/(0.2\sqrt{T})$ between -4 and 2.5, where SPX quotes are liquid;
- out-of-the-money options only: puts below the forward, calls above it.

The forward and discount factor are not assumed. For every strike quoted on both sides, put–call parity gives

$$C(K) - P(K) = D\,(F - K), \tag{1}$$

which is linear in K. A regression on near-the-money strikes gives the slope $-D$ and intercept DF. No stale spot, assumed dividend yield or external rate curve enters the surface.

Table 1: Data and audit summary

Item	Value
Raw option quotes (calls and puts)	4,930,870
Quote dates	311
Distinct expiries	378
Two-sided share of raw quotes	94.7%
Duplicate date–expiry–strike–type rows	0
Showcase date 2023-01-18: raw quotes → cleaned OTM surface	15,106 → 5,498 (44 expiries)
Calibration subset (8 tenors, 1 month to 2 years, 25 strikes each)	197
Weekly calibrations (2020 Q1 / 2023)	14 / 59

1.3 Models and notation

Under the risk-neutral measure with rate r and dividend yield q, the Heston model is

$$\frac{dS_t}{S_t} = (r-q)\,dt + \sqrt{v_t}\,dW_t^S, \qquad dv_t = \kappa(\theta - v_t)\,dt + \xi\sqrt{v_t}\,dW_t^v, \qquad d\langle W^S, W^v\rangle_t = \rho\,dt. \tag{2}$$

Its characteristic function is known in closed form; the code uses the "little Heston trap" formulation (Albrecher et al., 2007), which avoids the branch-cut discontinuity of the complex logarithm at long maturities. The COS method (Fang and Oosterlee, 2008) expands the density of $x = \ln(S_T/K)$ in a cosine series on a truncation interval $[a, b]$:

$$V_0 \approx e^{-rT} \sum_{k=0}^{N-1}{}' \operatorname{Re}\left[\varphi\left(\frac{k\pi}{b-a}\right) e^{-ik\pi\frac{a}{b-a}}\right] V_k, \tag{3}$$

where φ is the characteristic function, V_k are the cosine coefficients of the payoff (closed form for calls and puts), and the prime halves the first term. The interval comes from the cumulants of x, which are computed numerically from the cumulant generating function.

Local volatility is obtained from total implied variance $w(y,T) = \sigma_{\text{imp}}^2 T$ at log-moneyness $y = \ln(K/F)$ (Gatheral, 2006):

$$\sigma_{\text{loc}}^2(y,T) = \frac{\partial_T w}{1 - \frac{y}{w}\partial_y w + \frac{1}{4}\left(-\frac{1}{4} - \frac{1}{w} + \frac{y^2}{w^2}\right)(\partial_y w)^2 + \frac{1}{2}\partial_{yy} w}. \tag{4}$$

Each expiry is first fitted with an SVI slice (Gatheral and Jacquier, 2014), so the derivatives in the formula are analytic in y; the T-derivative uses linear interpolation in total variance between slices.

1.4 Validation protocol

Every reported number has an independent check:

- **Prices:** a second engine of a different type (Fourier vs PDE vs simulation), and a third where possible.
- **Monte Carlo:** agreement is judged in standard errors, never in decimals.
- **Convergence:** log–log slopes against the grid size or path count, compared with theory.
- **Calibration:** parameter recovery on synthetic surfaces, with and without added quote noise, before any SPX fit.
- **Hedging:** the simulated hedging error is compared with the Kamal–Derman formula before attributing real P&L.

2 Three Engines, One Price

2.1 Cross-validation table

Table 2 prices three contracts with every applicable engine. Black-Scholes and Heston use $S = 100$, $K = 95$, $T = 0.75$, $r = 4\%$, $q = 1.5\%$; the Heston parameters are those of Fang and Oosterlee (2008) ($v_0 = 0.0175$, $\kappa = 1.58$, $\theta = 0.0398$, $\xi = 0.575$, $\rho = -0.571$), which violate the Feller condition, so the variance process touches zero. This is the hard case for simulation, handled by the Andersen quadratic-exponential scheme (Andersen, 2008).

Table 2: Cross-validation of pricing engines

Product	Closed form	COS	Crank–Nicolson	Monte Carlo ($\pm$1 s.e.)
European call, Black-Scholes	12.0056	12.0056	12.0057	12.0042 $\pm$ 0.0058
European call, Heston	–	5.7852	–	5.7838 $\pm$ 0.0071
American put ($S = 36$, $K = 40$)	–	–	4.4867	4.4758 $\pm$ 0.0065

The American put uses the Longstaff and Schwartz (2001) test case. The PDE handles early exercise with a penalty method; Longstaff–Schwartz uses a Laguerre basis on 50 exercise dates, with the exercise rule fitted on one set of paths and applied to another. An out-of-sample rule is sub-optimal by construction, so its value is a *lower* bound for the 50-date Bermudan, which is itself below the American. The PDE value sits above the Monte Carlo value, as it should.

> **Published benchmarks have errors too.** The finite-difference value 4.478 quoted for this American put in Longstaff and Schwartz (2001) is off by 0.009. A 20,000-step binomial tree confirms 4.4867, the Crank–Nicolson value. Without a second engine this discrepancy would have looked like a bug in the PDE.

2.2 Convergence rates

Crank–Nicolson is second order in both space and time, but only if the payoff kink does not pollute the solution. Two details matter. First, the payoff is averaged over the grid cell that contains the strike. Second, the first steps use fully implicit half-steps (Rannacher start-up, Rannacher, 1984). The Crank–Nicolson amplification factor tends to -1 for stiff modes, so the high-frequency error created by the kink is never damped; implicit steps damp it immediately.

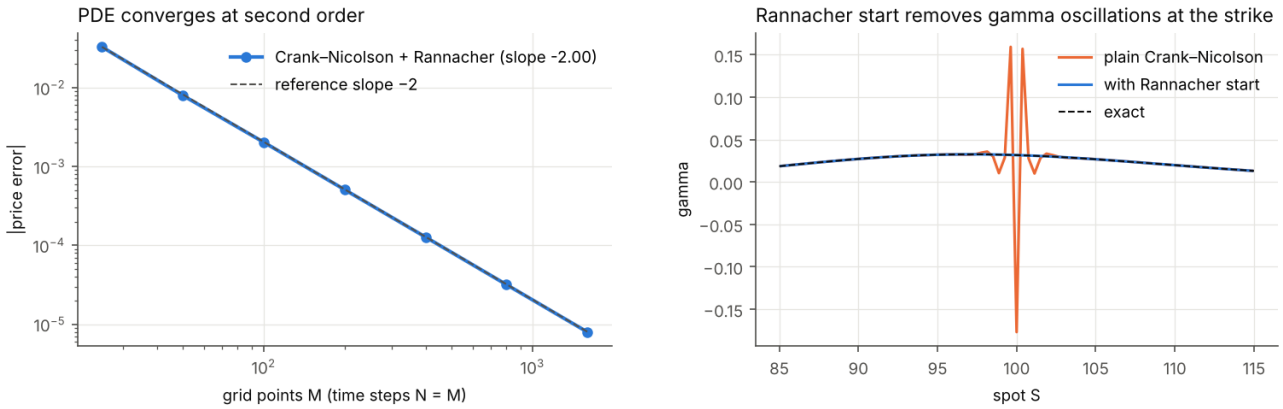


Figure 1: Left: the Crank–Nicolson price error falls with slope -2.00 when space and time are refined together. Right: with 25 time steps, plain Crank–Nicolson leaves a gamma spike of about 0.2 at the strike; four implicit half-steps remove it (error 10^{-5}).

Two further numerical lessons came from the exotics. A knock-out barrier that falls between grid nodes reduces the PDE to first order; aligning the grid so that the barrier is exactly a node restores second order. For the autocallable, the

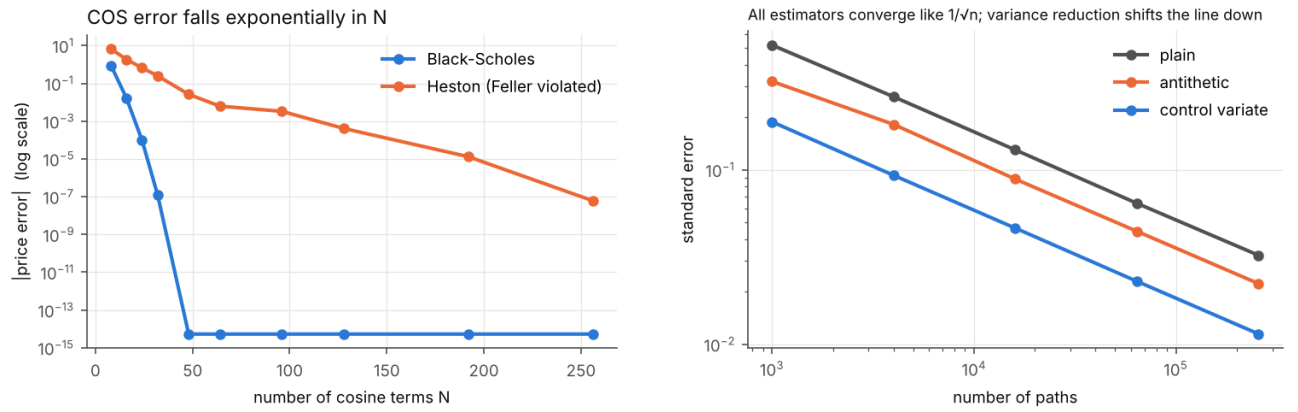


Figure 2: Left: COS converges exponentially in the number of terms; Black-Scholes reaches machine precision by $N \approx 64$, Heston with the Feller condition violated needs more terms because its density is less smooth. Right: all Monte Carlo estimators converge as $n^{-1/2}$; antithetic sampling cuts the variance 2.1× and the control variate $e^{-rT}S_T$ cuts it 8.0×.

value jumps at the autocall level and at the knock-in barrier; giving the node on each discontinuity the average of its two one-sided values turns first-order convergence into near-exact agreement with Monte Carlo by 300 space nodes.

3 Calibrating to the SPX Surface

3.1 Objective

The calibration minimises vega-weighted price errors over the 197 options in the calibration subset:

$$\min_{\Theta} \sum_{i} \left(\frac{C_i^{\text{model}}(\Theta) - C_i^{\text{mkt}}}{\max(\mathcal{V}_i,\, \underline{\mathcal{V}})} \right)^2, \tag{5}$$

where $\mathcal{V}_i$ is the Black-Scholes vega and the floor $\underline{\mathcal{V}}$ is 1% of the median vega. To first order each term is an implied-vol error, so the objective is measured in vol points while avoiding a root-finding step inside the optimiser. A bounded trust-region least-squares solver runs from several starting points; on the showcase date four of four starts reach the same minimum.

Before the SPX fit, the calibrator was run on synthetic Heston surfaces. It recovers the true parameters to 10^{-4}, and with 0.2 vol points of quote noise added the fit stops at the noise level (RMSE 0.16 vol pt).

3.2 One date in detail: 2023-01-18

Table 3: Fit quality on 2023-01-18 (implied-vol RMSE)

Expiry set	Heston (5 parameters, whole surface)	SABR (3 per expiry, $\beta = 1$)
All expiries	0.89 vol pt	0.43 vol pt
1 month	1.48 vol pt	0.54 vol pt
2 years	0.49 vol pt	0.35 vol pt

The fitted Heston parameters are $v_0 = 0.039$, $\kappa = 1.89$, $\theta = 0.072$, $\xi = 0.97$, $\rho = -0.71$, with Feller ratio $2\kappa\theta/\xi^2 = 0.29$. The Feller condition is violated, which is typical for equity index surfaces: the steep skew requires a vol of vol that the

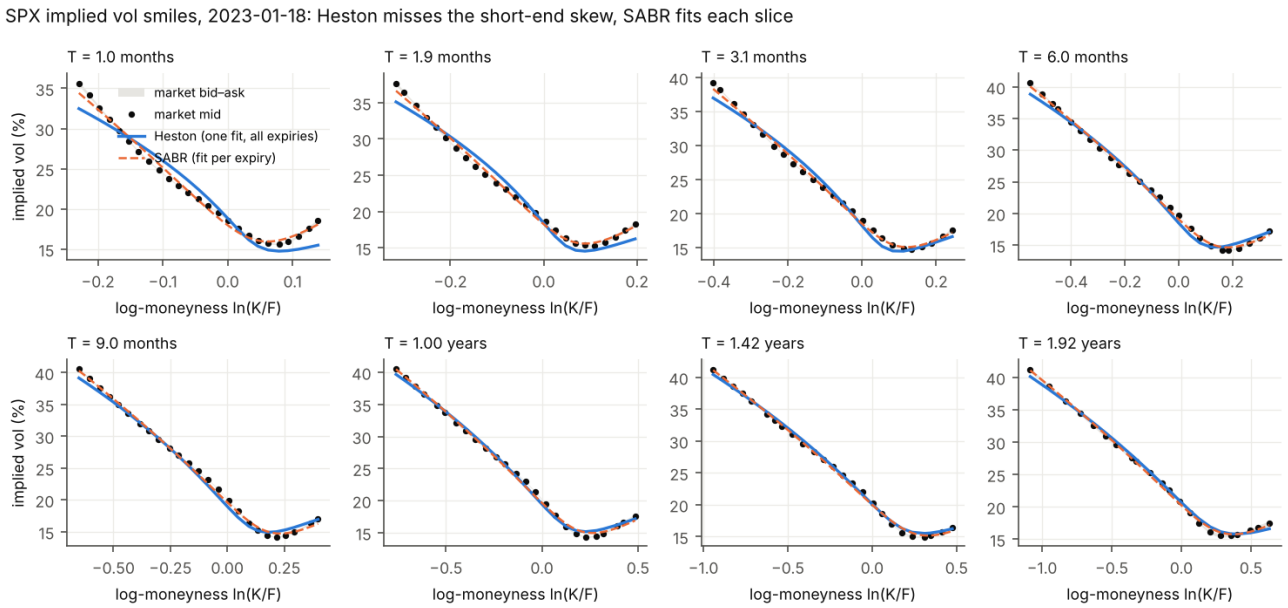


Figure 3: SPX implied vol smiles on 2023-01-18 with the market bid–ask band. One Heston parameter set (all expiries) misses the short-dated skew; SABR, refitted on every expiry, follows each slice.

condition does not allow.

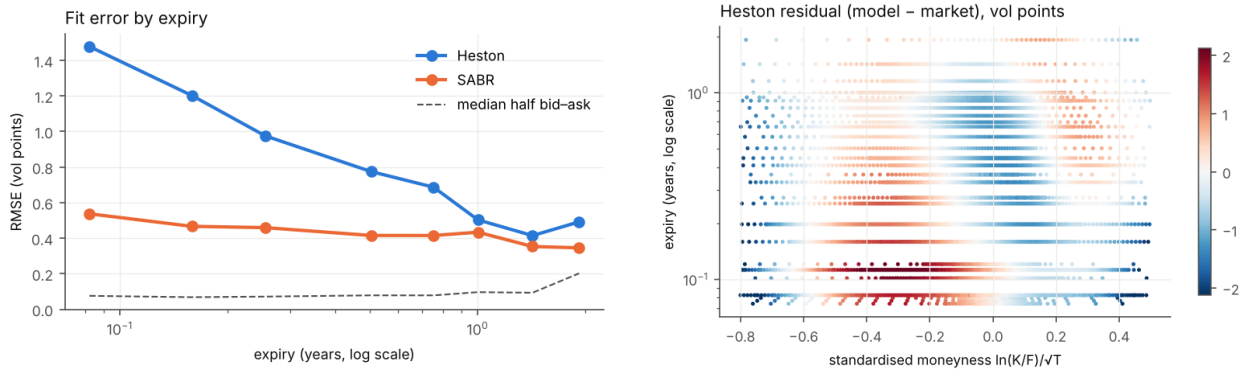


Figure 4: Left: Heston's fit error falls steadily with maturity; SABR's is flat. Both stay above the median half bid–ask spread. Right: Heston residuals are structured. At short maturities the model is too low in the deep put wing and too high around the money.

> **A model error, not a fitting failure.** In Heston the short-dated skew comes only from the correlated variance diffusion, which needs time to act. The market's 1-month skew is steeper than a diffusive stochastic-vol model can produce without extreme vol of vol. That is the standard motivation for jumps (Bates) or rough volatility, and the residual map shows exactly that signature.

SABR's per-expiry fit is better for a different reason. Its vol of vol ν decays from 2.25 at one month to 0.58 at two years, roughly as $T^{-0.45}$. The term structure is absorbed into parameters that change with maturity, so SABR is an interpolator of each slice rather than a dynamic model of the surface.

3.3 Local volatility

Dupire local vol built from the SVI slices reproduces the vanilla surface by construction (verified on synthetic data to 0.03 vol pt). Its at-the-money skew is 1.95 times the implied skew, close to the short-maturity rule that implied vol at strike K is roughly the average of local vol between spot and K, so the implied slope is half the local slope.



Figure 5: Market implied vol, SVI fit and Dupire local vol at $T = 0.5$ years on 2023-01-18. The local-vol skew is about twice the implied skew.

4 Parameter Stability and Identifiability

4.1 What the surface actually identifies

A good fit on one date says nothing about whether the parameters mean anything. The project therefore asked the identifiability question twice: first on synthetic data, where the answer is known, and then on 73 weekly SPX calibrations.

On synthetic surfaces a κ profile (fix κ, recalibrate the other four parameters) is flat around the true value when the surface contains only short expiries: ξ moves to compensate, and the fit barely changes. The prediction was therefore that on real data κ and ξ would not be separately pinned down.

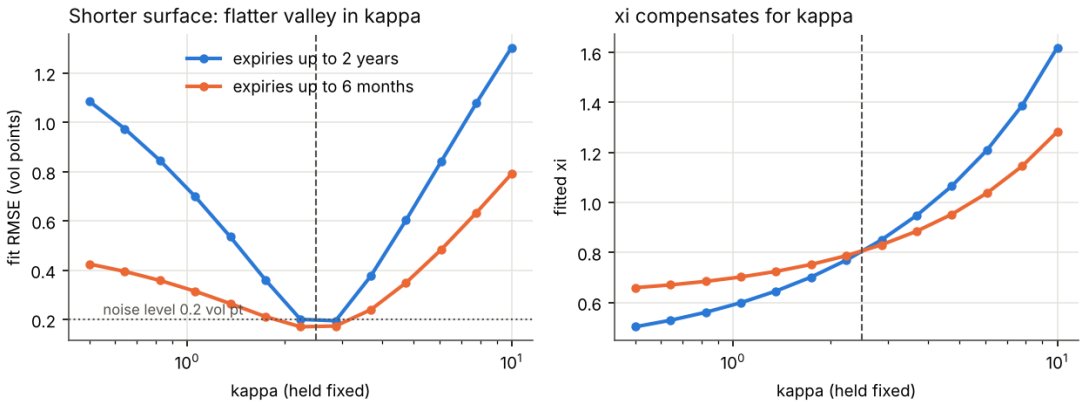


Figure 6: Synthetic Heston surface. Left: the fit RMSE as a function of a fixed κ has a flat valley, flatter when only expiries up to 6 months are used. Right: the fitted ξ rises with κ along the valley.

4.2 Weekly SPX calibrations

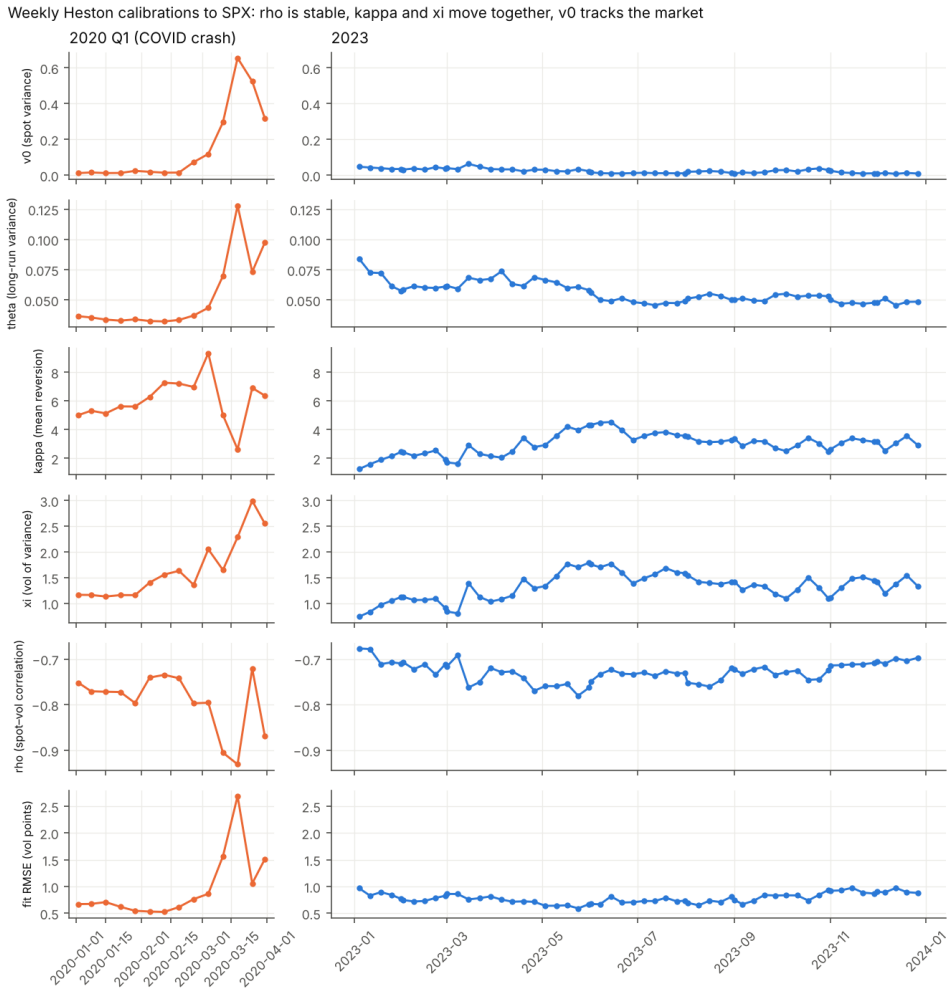


Figure 7: Weekly Heston calibrations to SPX. ρ is stable, κ and ξ move together, and v_0 tracks short-dated implied vol. During the COVID crash every parameter and the fit error move sharply.

Table 4: Heston parameter stability, 59 weekly fits in 2023

Parameter	Mean	Coefficient of variation	Corr. of weekly changes with $\Delta\kappa$
ρ	-0.73	3%	-0.23
θ	0.056	15%	-0.27
ξ	1.32	20%	**0.96**
κ	2.99	25%	1
v_0	0.024	54%	0.05
Fit RMSE	0.78 vol pt	12%	

- ρ **is the stable parameter.** The SPX skew is a persistent feature of the market.
- κ **and** ξ **move together.** Their week-on-week changes have correlation 0.96, the ridge predicted on synthetic data. The surface pins down a combination of the two, not each separately.
- v_0 **is a state variable, not a parameter.** It should move, and it does: $\sqrt{v_0}$ has correlation 0.99 with 1-month at-the-money implied vol.
- **The crisis breaks the fit.** From 2020-02-19 to 2020-03-18, $\sqrt{v_0}$ rises from 12% to 81%, ρ falls to -0.93 and the fit RMSE rises from below 1 to 2.7 vol pt.

Practical consequence. Any risk number that depends on κ alone, such as sensitivity to the long-run variance, is unreliable, and recalibrating daily injects noise into P&L. Two remedies follow: fix κ and recalibrate the rest, or penalise changes from the previous day's parameters. A five-parameter diffusion fits a calm surface well and a crisis surface badly, exactly when the model matters most.

5 Exotics and Model Risk

5.1 Products and their checks

Down-and-in barrier put. The continuously monitored price has a closed form (Reiner and Rubinstein, 1991). Daily monitoring is handled by the Broadie et al. (1997) correction, which shifts the barrier away from spot by $e^{-\beta\sigma\sqrt{\Delta t}}$ with $\beta = -\zeta(1/2)/\sqrt{2\pi} \approx 0.5826$. The shifted closed form gives 4.92% of spot for the 3-year 100/60 put; daily-monitored Monte Carlo gives 4.93 ± 0.04.

Autocallable. A 3-year note on SPX with notional 100 and quarterly observations. If the index is at or above its initial level on an observation date, the note redeems at par plus an 8% p.a. accrued coupon. If it is never called and the index has ever closed below 60% (daily knock-in), the holder receives $100\, S_T/S_0$ when $S_T < S_0$; otherwise par. The note is priced by Monte Carlo and by two coupled PDEs ("knocked in" and "not knocked in"), stepped once per business day. Under Black-Scholes the PDE gives 97.18 against 97.20 ± 0.04 by Monte Carlo; under local vol both give 96.63.

The present value is affine in the coupon, $PV(c) = A + cB$, so two pricings give the fair coupon $c^* = (100 - A)/B$ at which the note is worth par.

5.2 Same market, three models

All three models are fitted to the 2023-01-18 SPX surface (spot 3945.1, implied $r = 3.93\%$, $q = 1.49\%$). Black-Scholes uses the market at-the-money vol at the note's maturity, 20.65%.

Table 5: Autocallable and barrier put under three models fitted to the same surface

Model	Fair coupon	P(called, 1st date)	P(knock-in loss)	Exp. life (y)	DIP 100/60 (% S_0)
Black-Scholes	14.03%	50.3%	8.2%	0.96	4.93
Heston	15.17%	63.8%	8.3%	0.73	9.32
Local vol	15.95%	58.8%	8.9%	0.73	8.43

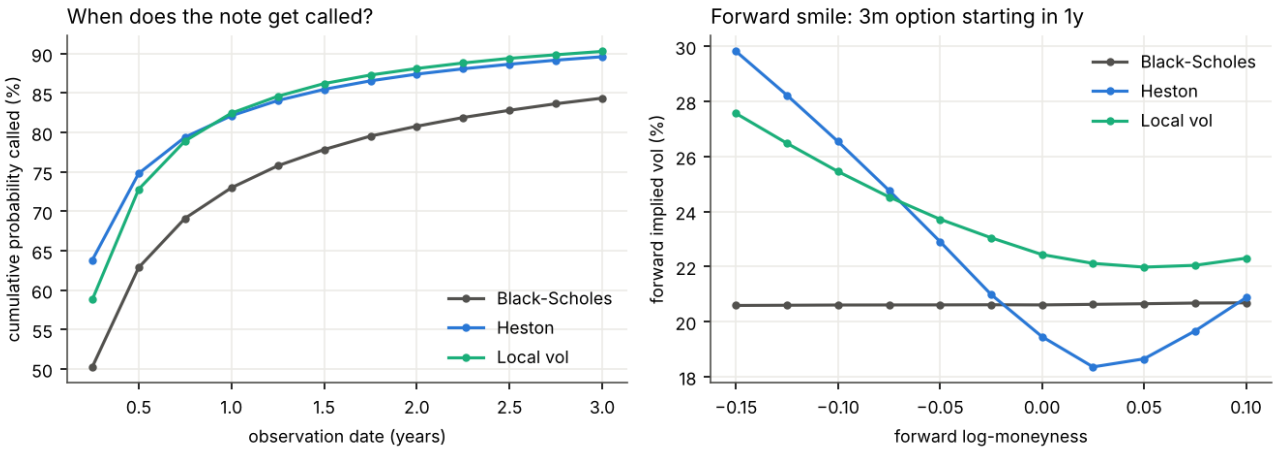


Figure 8: Left: cumulative probability that the note has been called by each observation date. Right: forward smile, the implied vol of a 3-month option starting in one year, computed from each model's simulated paths. Local vol's forward skew is flatter than Heston's.

> **Economic interpretation.** Black-Scholes, with no skew, values the 60% knock-in put at about half what the skew-consistent models do, and so understates the coupon the issuer can afford. Heston and local vol both reproduce the vanilla market, yet differ by 0.8 coupon points. The difference is the forward smile: local vol's skew flattens as time passes, Heston's persists. A path-dependent payoff depends on those forward dynamics, which vanilla prices do not pin down.

The industry answer is stochastic local vol, which matches vanillas exactly and takes its forward-smile dynamics from a stochastic component. That moves the model risk into a single mixing weight, which then becomes the parameter a desk must reserve against.

6 Greeks and the Sources of Hedging Error

6.1 Greeks without bumping

Monte Carlo Greeks were implemented three ways. The pathwise estimator differentiates along each path, $\Delta = \mathbb{E}[e^{-rT}\mathbf{1}\{S_T > K\}S_T/S_0]$, and fails for a digital option, whose payoff has zero derivative almost everywhere. The likelihood-ratio estimator moves the derivative onto the density, $\Delta = \mathbb{E}[e^{-rT} f(S_T)\, Z/(S_0\sigma\sqrt{T})]$, and works for any payoff at the cost of higher variance. A mixed estimator, likelihood ratio applied to the pathwise delta, gives gamma. All three match the closed forms within three standard errors.

6.2 The hedged P&L identity

For a short option that is delta-hedged and self-financed, the Black-Scholes PDE turns theta and financing into $\frac{1}{2}\Gamma S^2\sigma_{\text{imp}}^2\,dt$, so one day's P&L is, to second order,

$$d\Pi \approx \underbrace{\tfrac{1}{2}\Gamma S^2\left(\sigma_{\text{imp}}^2\,dt - \left(\tfrac{dS}{S}\right)^2\right)}_{\text{gamma/theta}} - \underbrace{\mathcal{V}\,d\sigma_{\text{imp}}}_{\text{vega}} - \underbrace{\left(\text{vanna}\cdot dS\,d\sigma_{\text{imp}} + \tfrac{1}{2}\text{volga}\cdot d\sigma_{\text{imp}}^2\right)}_{\text{vanna/volga}} + \varepsilon. \tag{6}$$

The first term is the core of option trading: a short-gamma book earns when realised variance is below implied and loses when it is above.

6.3 Simulation: what discrete hedging costs

In a Black-Scholes world with a 3-month at-the-money call and $\sigma = 20\%$, the hedging error from rebalancing N times should follow Kamal and Derman (1999):

$$\text{sd}(\text{P\&L}) \approx \sqrt{\frac{\pi}{4}}\,\frac{\mathcal{V}\,\sigma}{\sqrt{N}}. \tag{7}$$

With 63 daily rebalances the simulated standard deviation is 0.434 against 0.441 from the formula (per 100 notional). Halving the error needs four times the trades. With 5 bp transaction costs, cost grows like $\sqrt{N}$ while hedging error falls like $1/\sqrt{N}$, so there is an optimal frequency, near 252 rebalances here.

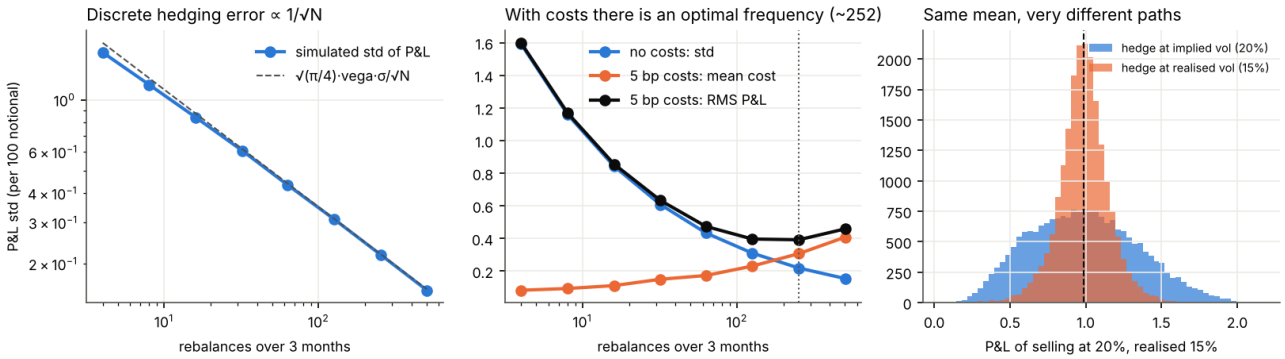


Figure 9: Left: discrete-hedging error scales as $N^{-1/2}$ and matches Kamal–Derman. Centre: with 5 bp costs the RMS P&L has a minimum. Right: selling at 20% implied when realised is 15%; hedging with implied-vol deltas and with realised-vol deltas gives the same mean (0.99) but very different dispersion (0.35 vs 0.16).

6.4 Real SPX: short 1-month calls, hedged daily

On the first trading day of each month, one 1-month at-the-money SPX call is sold at mid (strike nearest the implied forward, expiry nearest 30 days), hedged daily in the index, and re-marked daily at market mid. Each day's P&L is split with the identity above using that day's implied vols.

Table 6: Real SPX hedging backtest, index points per one call, no transaction costs

Period	Trades	P&L	Gamma/theta	Vega	Vanna/volga	Residual	Entry IV	Realised
2020 Q1	3	−127.8	−96.4	−65.5	10.6	23.5	17.2%	36.3%
2023	12	+31.8	+73.6	−75.6	2.1	31.7	15.6%	12.1%

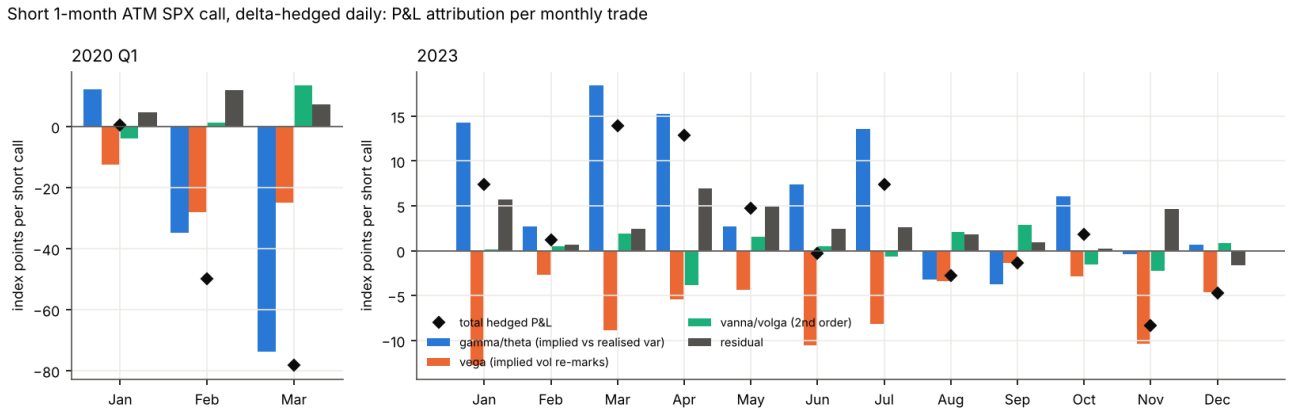


Figure 10: P&L attribution per monthly trade. In 2023 gamma/theta (blue) is mostly positive, and most of it is given back through vega (orange). In 2020 Q1 both lines are negative.

- **2023 is the variance risk premium at work.** Implied vol averaged 15.6% against 12.1% realised, and gamma/theta earned $+73.6$ points over 12 trades. Most of it was given back through vega: in a rising market a fixed strike slides down the skew, so its implied vol rises even when the smile itself does not move.
- **2020 Q1 is the tail of selling volatility.** Realised vol of 36% against 17% implied cost 96.4 points in gamma/theta in three trades.
- **The Greeks explain most of the P&L.** Gamma/theta and vega explain 86% of the variance of daily hedged P&L; adding vanna and volga raises this to 91%.

> **Inference discipline.** Fifteen trades in two hand-picked windows cannot estimate the return of selling volatility, and the March 2020 trade stops after eight days at the data cut-off. The backtest is evidence about *where* hedged P&L comes from, not about how much a strategy would earn. The positive residual, concentrated on the largest down days, is still unexplained; third-order spot terms and bid–ask noise in daily mids are the two candidates.

7 Evidence Ladder

The project supports five claims of different strength.

1. **Numerical correctness: strong.** Three engine types agree on every contract within discretisation or statistical error, and each converges at its theoretical rate.
2. **Heston's short-end misfit is a model limitation: strong.** The calibrator recovers synthetic parameters to 10^{-4}, all starts agree, and the residuals have the structure diffusive stochastic vol predicts.
3. **κ–ξ non-identifiability: strong.** Predicted on synthetic data before the SPX fits, then observed in 59 weekly fits (correlation 0.96).
4. **Model risk on path-dependent payoffs: strong for the mechanism, single-date for the size.** The forward-smile explanation is shown directly; the 2-point coupon range is from one surface.
5. **P&L attribution on real data: supportive.** The Greeks explain 91% of daily P&L variance, but over 15 trades only.

The contribution is therefore not a new model. It is a pricing library in which every number is checked by an independent method, so that the differences that remain, between models and between hedge and reality, can be attributed to their economic cause.

8 Limitations & Next Tests

8.1 Current limitations

- **Data:** end-of-day mids from a single vendor, without intraday synchronisation between the index and the options; constant r and q per date when building local vol.
- **Heston PDE:** there is no 2D (ADI) solver, so Heston prices are cross-checked by COS and Monte Carlo only.
- **Models:** no jumps and no rough volatility, which the short-dated residuals call for.
- **Model risk:** the coupon comparison uses one surface; its stability over time is untested.
- **Trading frictions:** the real backtest sells at mid and hedges without cost.

8.2 Highest-value next tests

1. Stochastic local vol for the autocallable, with the mixing weight as an explicit model-risk parameter.
2. Andersen–Broadie dual upper bounds to bracket the Longstaff–Schwartz value from above.
3. A 2D Crank–Nicolson ADI solver for Heston, as a third independent engine.
4. A smile-adjusted (minimum-variance) delta in the backtest, to move the sticky-strike vega loss into the hedge.
5. Bid–ask and index-futures costs in the real backtest, and a full-revaluation P&L explain to identify the residual.

9 Closing View

The project starts with a narrow requirement: the same contract must get the same price from three unrelated methods. Once that holds, the remaining disagreements become informative. Heston's 1-month error says that diffusion is not enough at the short end. The κ–ξ ridge says that a good fit can still leave parameters unidentified. The 2-point coupon gap says that matching vanillas does not fix the price of path dependence. The hedging backtest says that the Black-Scholes Greeks account for most, but not all, of what a hedged book earns.

Each of these is a statement about a model, not about the code. That is the purpose of the cross-validation.

A Reproducibility Checklist

- 70 unit tests (pytest) run in continuous integration on every push; they use synthetic inputs only and need no market data.
- Black-Scholes: textbook values, put–call parity, no-arbitrage bounds, Greeks against finite differences, and Greeks satisfying the Black-Scholes PDE.
- Heston COS: agreement with the Lewis (2001) integral by adaptive quadrature to 10^{-9}, and the Black-Scholes limit as $\xi \to 0$ at rate $O(\xi^2)$.
- PDE: convergence order on Europeans, American put against a binomial tree, barrier order with the grid aligned.
- Monte Carlo: every estimator within three standard errors of the analytic value; Longstaff–Schwartz below the PDE American value.
- Calibration: implied forward recovered from synthetic chains; Heston parameter recovery with and without noise; Dupire round trip on a synthetic surface to 0.03 vol pt.
- Exotics: barrier closed form against PDE and Monte Carlo; autocallable PDE against Monte Carlo.

- Hedging: Kamal–Derman scaling and the hedged P&L identity on simulated paths.
- Every formula in the code is derived in `docs/derivations/01–10`; each stage script regenerates its tables and figures.

References

Hansjörg Albrecher, Philipp Mayer, Wim Schoutens, and Jurgen Tistaert. The little Heston trap. *Wilmott Magazine*, pages 83–92, January 2007.

Leif Andersen. Simple and efficient simulation of the Heston stochastic volatility model. *Journal of Computational Finance*, 11(3):1–42, 2008.

Mark Broadie, Paul Glasserman, and Steven Kou. A continuity correction for discrete barrier options. *Mathematical Finance*, 7(4):325–349, 1997.

Bruno Dupire. Pricing with a smile. *Risk*, 7(1):18–20, 1994.

Fang Fang and Cornelis W. Oosterlee. A novel pricing method for European options based on Fourier-cosine series expansions. *SIAM Journal on Scientific Computing*, 31(2):826–848, 2008.

Jim Gatheral. *The Volatility Surface: A Practitioner's Guide*. Wiley, 2006.

Jim Gatheral and Antoine Jacquier. Arbitrage-free SVI volatility surfaces. *Quantitative Finance*, 14(1):59–71, 2014.

Patrick S. Hagan, Deep Kumar, Andrew S. Lesniewski, and Diana E. Woodward. Managing smile risk. *Wilmott Magazine*, pages 84–108, September 2002.

Steven L. Heston. A closed-form solution for options with stochastic volatility with applications to bond and currency options. *The Review of Financial Studies*, 6(2):327–343, 1993.

Michael Kamal and Emanuel Derman. Correcting Black-Scholes. *Risk*, 12(1):82–85, 1999.

Alan L. Lewis. A simple option formula for general jump-diffusion and other exponential Lévy processes. Working paper, OptionCity.net, 2001.

Francis A. Longstaff and Eduardo S. Schwartz. Valuing American options by simulation: a simple least-squares approach. *The Review of Financial Studies*, 14(1):113–147, 2001.

Rolf Rannacher. Finite element solution of diffusion problems with irregular data. *Numerische Mathematik*, 43(2):309–327, 1984.

Eric Reiner and Mark Rubinstein. Breaking down the barriers. *Risk*, 4(8):28–35, 1991.